

VAL-D'OR, QC, Canada

WMO: 717250

Lat: 48.0564N

Lon: 77.7867W

Elev: 1107

StdP: 14.12

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.3	-20.6	-34.3	0.8	-26.1	-28.6	1.2	-20.3	22.4	14.7	20.6	15.1	4.5	290	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	83.9	67.2	80.9	65.4	77.9	64.0	70.5	79.1	68.5	76.1	66.7	74.0	10.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	106.3	73.9	65.9	99.4	72.3	63.9	92.7	70.4	35.1	79.5	33.5	76.0	31.9	74.1	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.7	18.4	16.4	DB	-34.1	89.8	5.4	2.7	-38.0	91.8	-41.1	93.4	-44.2	94.9	-48.1	96.9	(4)
			WB	-34.3	74.0	5.4	2.4	-38.1	75.7	-41.3	77.1	-44.3	78.4	-48.2	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	3.6	7.1	18.8	33.9	49.6	60.0	64.4	61.6	54.1	41.4	27.1	12.9	(6)
(7)		DBStd	23.61	14.58	13.03	13.22	10.10	9.46	8.11	6.03	6.34	8.31	8.74	10.69	13.17	(7)
(8)		HDD50	6419	1439	1202	969	487	126	12	0	1	49	296	687	1151	(8)
(9)		HDD65	10620	1904	1622	1433	932	484	190	86	145	340	732	1136	1616	(9)
(10)		CDD50	1443	0	0	1	5	114	314	446	361	172	29	1	0	(10)
(11)		CDD65	168	0	0	0	0	8	41	67	39	13	0	0	0	(11)
(12)		CDH74	1638	0	0	1	2	123	464	608	333	105	2	0	0	(12)
(13)		CDH80	356	0	0	0	0	28	113	141	59	15	0	0	0	(13)
(14)	Wind	WSAvg	7.9	7.7	8.0	8.6	8.8	8.3	7.6	7.0	6.7	7.4	8.1	8.3	7.8	(14)
(15)	Precipitation	PrecAvg	38.6	2.4	2.1	2.7	2.6	3.0	3.9	4.0	4.1	4.1	3.4	3.4	3.0	(15)
(16)		PrecMax	47.9	5.6	4.4	5.4	5.0	6.0	7.7	8.1	8.9	7.5	7.4	6.8	4.6	(16)
(17)		PrecMin	30.1	0.9	0.6	0.6	1.1	0.5	1.2	1.3	1.6	2.5	1.3	1.1	1.6	(17)
(18)		PrecStd	4.6	1.0	1.0	1.1	1.0	1.3	1.3	1.6	1.7	1.3	1.3	1.4	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	41.1	56.4	68.6	84.1	87.8	89.0	85.2	82.0	72.0	55.6	42.0	(19)
(20)		MCWB	40.9	37.6	48.7	53.5	63.6	67.9	70.3	68.5	67.3	61.0	51.2	40.1	(20)	
(21)		2%	DB	34.5	35.4	47.1	60.4	77.0	83.2	83.7	81.3	77.1	65.3	49.6	36.7	(21)
(22)		MCWB	33.7	33.3	40.1	46.6	59.4	65.8	67.6	66.7	65.2	56.9	46.7	35.7	(22)	
(23)		5%	DB	30.6	31.8	42.0	54.7	72.4	79.5	80.4	77.9	72.6	60.0	45.0	33.7	(23)
(24)		MCWB	29.8	30.2	36.5	44.6	57.3	63.6	65.7	65.1	63.1	54.6	42.7	32.9	(24)	
(25)		10%	DB	26.2	27.8	37.5	49.2	67.1	75.5	77.1	74.2	68.5	55.4	41.0	31.7	(25)
(26)		MCWB	25.3	26.3	33.5	41.1	54.4	61.6	64.2	63.5	61.4	51.0	38.8	31.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.7	38.8	50.8	56.6	68.3	72.0	73.8	71.7	70.0	62.7	52.6	40.7	(27)
(28)		MCDWB	40.9	40.2	55.1	66.1	78.1	82.8	83.2	80.9	77.3	68.7	54.0	41.4	(28)	
(29)		2%	WB	33.8	33.2	41.3	49.5	62.8	69.1	70.7	69.3	67.7	58.3	47.2	35.8	(29)
(30)		MCDWB	34.3	34.9	45.4	56.3	72.9	78.6	78.5	77.1	73.9	63.5	48.8	36.6	(30)	
(31)		5%	WB	29.9	30.2	37.0	45.3	59.6	66.6	68.4	67.2	65.1	55.1	42.9	33.2	(31)
(32)		MCDWB	30.6	31.7	41.0	53.1	67.9	74.3	75.7	74.2	71.1	59.6	44.7	33.7	(32)	
(33)		10%	WB	25.3	26.3	33.6	41.8	55.9	64.2	66.4	65.1	62.1	51.6	39.0	30.9	(33)
(34)		MCDWB	26.3	27.8	37.1	48.6	64.7	71.7	74.1	71.8	67.6	55.0	40.6	31.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	19.4	21.0	21.0	19.3	22.2	21.8	20.0	19.3	18.7	14.3	12.7	15.6	(35)	
(36)		5% DB	MCDBR	17.5	18.5	22.4	28.3	29.3	26.5	24.3	23.8	23.4	22.1	16.8	13.0	(36)
(37)		MCWBR	17.3	17.1	16.5	17.6	14.9	12.5	11.1	11.7	12.9	15.2	14.2	12.8	(37)	
(38)		5% WB	MCDBR	17.9	18.6	20.0	25.1	24.4	21.4	20.0	19.8	20.1	20.3	15.7	12.8	(38)
(39)		MCWBR	17.9	17.6	15.8	17.4	14.3	12.2	11.1	11.3	12.6	15.4	14.2	12.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.272	0.280	0.308	0.358	0.361	0.381	0.394	0.378	0.362	0.328	0.303	0.260	(40)	
(41)		taud	2.221	2.215	2.221	2.296	2.404	2.378	2.366	2.403	2.424	2.505	2.441	2.338	(41)	
(42)		Ebn at Noon	252	278	288	283	286	280	274	274	267	261	243	245	(42)	
(43)		Edh at Noon	25	31	37	38	36	37	37	34	30	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	405	729	1192	1478	1641	1787	1704	1429	1092	636	335	273	(44)	
(45)		RadStd	59	104	119	117	153	118	159	97	114	87	41	39	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page